Subject: Final Project Proposal - Numerov Method for Quantum Bound States

Shalom Omer,

This is Alon Sportes again from the "Numerical methods in astrophysics" course. Following our agreement below, I would like to propose a final project focused on solving the one-dimensional time-independent Schrödinger equation using the Numerov method.

The project will focus on building a numerical solver for bound states in different potentials by combining Numerov integration with a shooting method. The boundary value problem will be reformulated as a root-finding problem, where eigenvalues are determined using bracketing and bisection. I will mainly consider symmetric potentials, which allows the use of parity (even/odd states) to simplify the boundary conditions. The solver will be validated against analytical solutions where available and through convergence tests.

I plan to apply the method to several systems, including:

- Infinite square well and harmonic oscillator (validation cases)
- Finite square well and quartic double-well (nontrivial bound states and tunneling effects)
- Finite barrier and double-barrier potentials (to explore scattering and resonances)

I will also compare the Numerov method with an RK4 integrator for the harmonic oscillator to examine accuracy and efficiency.

Please let me know if this topic is approved, and I will proceed accordingly.

Best, Alon Sportes